FSD2 LAB • EXPERIMENT 09

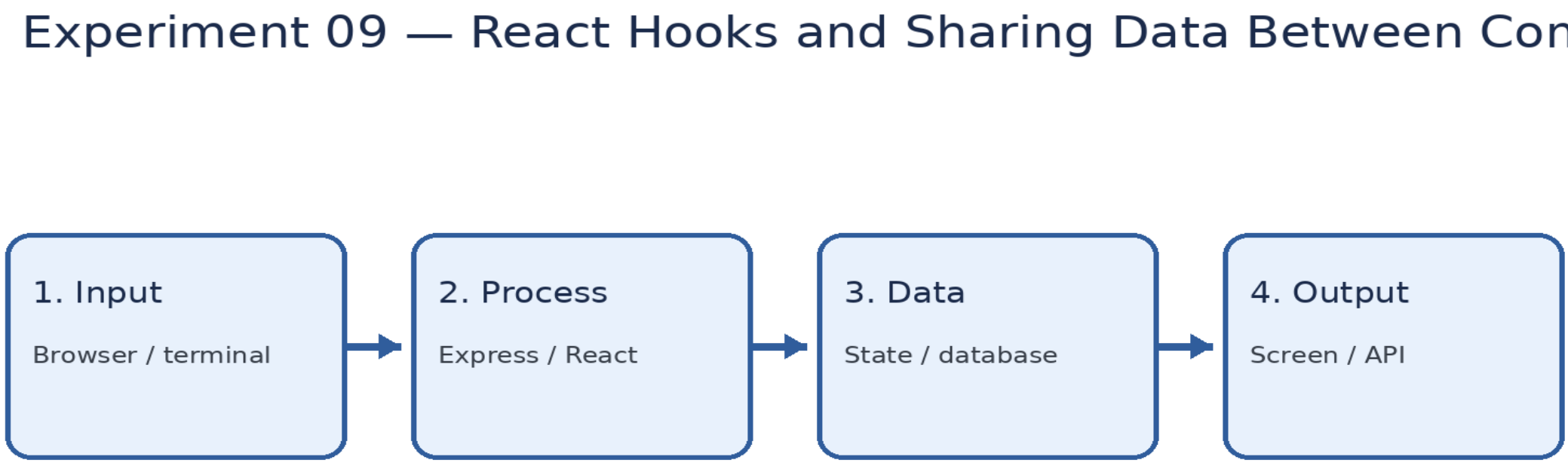
Experiment 09 — React Hooks and Sharing Data Between Components

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- `useState`: local state in a function component.
- Lifting state up: place shared state in the closest common parent and pass values/callbacks to children.
- `createContext` / `useContext`: share a value across the component tree.
- Class vs function component: two ways shown side-by-side to explain why hooks are useful.

2. Workflow



Workflow: input → processing → data/state → output

3.1. Code pathway — index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />

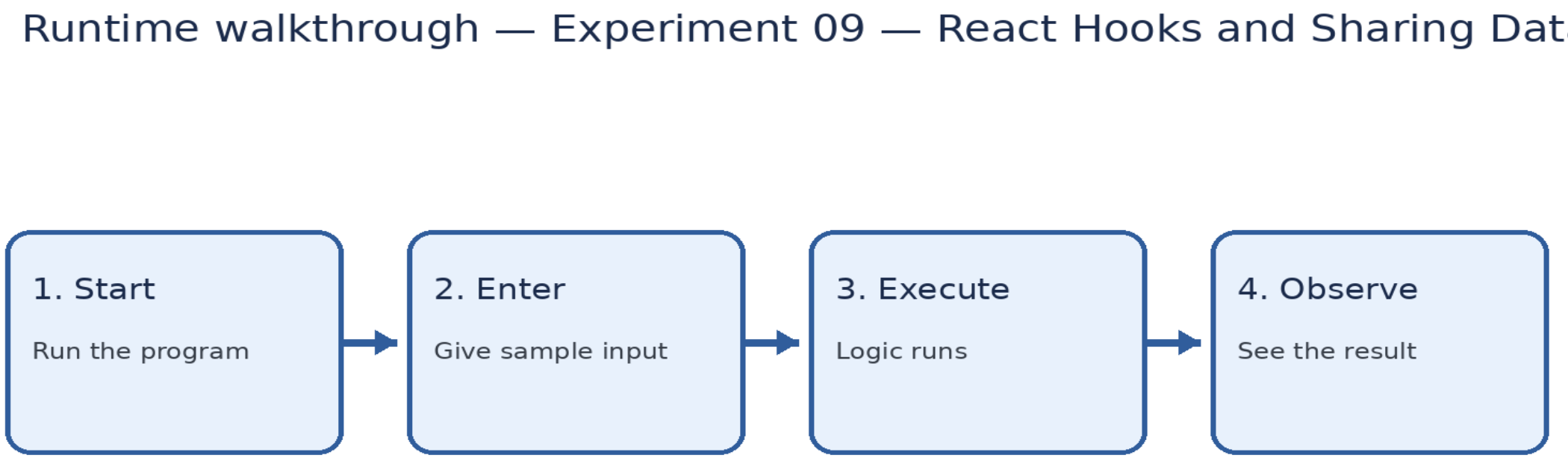
<meta
name="viewport"
content="width=device-width, initial-scale=1.0"
/>

```

3.2. Code pathway — index.html

```
11 <title>
12 FSD2 React Experiment 09
13 </title>
14 </head>
15
16 <body>
17 <div id="root"></div>
18
19 <script
20 type="module"
```

4. Runtime walkthrough



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?